

Assignment 2 SECPH 2523

QUESTION 1

Table 2.1 STUDENTFINE records data on traffic offences by students in a college.

TABLE 2.1 STUDENTFINE

StudentID	Name	LicenseNo	CarReg	ReceiptNo	Date	OffenceCode	Fine(RM)
F1011	CK Fong	P100	JGK9201	00-345	10/03/06	2	30.00
				00-690	15/03/06	1	20.00
				00-810	20/03/06	2	30.00
L782	AdamJames	P290	JJL99	00-459	13/03/06	1	20.00
H762	MalaDewi	P764	JJK7654	00-564	12/03/06	3	40.00
				00-770	19/06/06	2	30.00

a) Table 2.1 is said to be unnormalized table. Why?

Some cells have multiple values.

b) Transform the unnormalized Table 2.1 into a First Normal Form (1NF) table. Show the transformed 1 NF table. What is the primary key for this new 1 NF table.

The primary key: StudentID

TABLE 2.1 STUDENTFINE

StudentID	Name	LicenseNo	CarReg	ReceiptNo	Date	OffenceCode	Fine(RM)
F1011	CK FONG	P100	JKG9201	00-345	10/3/06	2	30.00
F1011	CK FONG	P100	JKG9201	00-690	15/03/06	1	20.00
F1011	CK FONG	P100	JKG9201	00-810	20/03/06	2	30.00
L782	AdamJames	P290	JJL99	00-459	13/03/06	1	20.00
H762	MalaDewi	P764	JJK76540	00-564	12/03/06	3	40.00
H762	MalaDewi	P764	JJK76540	00-77	19/06/06	2	30.00

QUESTION 2

The following table is un-normalized table that is created to manage a consulting company's operations. Create its normalized forms (1NF, 2NF, and 3NF).

Note: Define the PK and FK for each table:

Project number	Project name	Employee number	Employee name	Rate category	Hourly rate
1023	Madagascar travel site	11	Vincent Radebe	A	\$60
		12	Pauline James	B	\$50
		16	Charles Ramoraz	C	\$40
1056	Online estate agency	11	Vincent Radebe	A	\$60
		17	Monique Williams	B	\$50

1NF

Project number	Project Name	Employee number	Employee name	Rate category	Hourly rate
1023	Madagascar travel site	11	Vincent Radebe	A	\$60
1023	Madagascar travel site	12	Pauline James	B	\$50
1023	Madagascar travel site	16	Charles Ramoraz	C	\$40
1056	Online estate agency	11	Vincent Radebe	A	\$60
1056	Online estate agency	17	Monique Williams	B	\$50

PK: Project Number

2NF

Project number	Project Name
1023	Madagascar travel site
1023	Madagascar travel site
1023	Madagascar travel site
1056	Online estate agency
1056	Online estate agency

PK: Project Number

Employee number	Employee name	Rate category	Hourly rate
11	Vincent Radebe	A	\$60
12	Pauline James	B	\$50
16	Charles Ramoraz	C	\$40
11	Vincent Radebe	A	\$60
17	Monique Williams	B	\$50

Primary Key: Employee number

3NF

Project number	Project Name
1023	Madagascar travel site
1023	Madagascar travel site
1023	Madagascar travel site
1056	Online estate agency
1056	Online estate agency

Primary Key: Project Number

Project Number	Employee number	Rate
1023	11	A
1023	12	B
1023	16	C
1056	11	A
1056	17	B

Foreign key: Project number,Employee number,Rate category

Employee number	Employee name
11	Vincent Radebe
12	Pauline James
16	Charles Ramoraz
11	Vincent Radebe
17	Monique Williams

Primary Key: Employee number

Rate category	Hourly Rate
A	\$60
B	\$50
C	\$40
A	\$60
B	\$50

Primary Key: Rate category